

Grades will be notified at

7:00pm of Thursday, March 30, 2023

Question 1 (7.5 points)

Consider a country where only the goods market exists and in which $C = c_0$, $I = \bar{I} + d_1 Y$, $G = \bar{G}$, and $T = \bar{T} + tY$, where $c_0 > 0$, $0 < d_1 < 1$, $0 < t < 1$ and $t < d_1$. Starting from an initial equilibrium position, consumption falls by $\Delta c_0 < 0$. Compute the analytical expressions of the changes in equilibrium private saving and in equilibrium national saving caused by this decrease in c_0 .

SKETCH OF ANSWER

In this economy, equilibrium income is the only value of Y for which $Y = c_0 + \bar{I} + d_1 Y + \bar{G}$, or $\hat{Y} = (c_0 + \bar{I} + \bar{G})/(1 - d_1)$. Since in the initial equilibrium private saving is $S_{priv} = (1 - t)\hat{Y} - \bar{T} - c_0$, in the move from the initial to the new equilibrium private saving will fall by $\Delta S_{priv} = (1 - t)\Delta\hat{Y} - \Delta c_0 = [(1 - t)/(1 - d_1)]\Delta c_0 - \Delta c_0 = [(-t + d_1)/(1 - d_1)]\Delta c_0$. Since government saving $S_{gov} = T - G$ changes by $\Delta S_{gov} = t\Delta\hat{Y}$, the change in national saving – the sum of private and government saving – will be $\Delta S_{priv} + \Delta S_{gov} = (1 - t)\Delta\hat{Y} - \Delta c_0 + t\Delta\hat{Y} = \Delta\hat{Y} - \Delta c_0 = [d_1/(1 - d_1)]\Delta c_0$. [Fully equivalent, and faster, way to reach the same conclusion: since, in equilibrium, national saving equals investment, it will have to change by $\Delta I = d_1\Delta\hat{Y} = [d_1/(1 - d_1)]\Delta c_0$].

Question 2 (7.5 points)

True or false?

Motivate your answer in a brief but rigorous way, making explicit reference to the relevant theory. Lack of proper explanations will result in zero points.

“If the parameter c (the fraction of their money individuals want to hold in the form of currency) decreases, the interest rate for which the money market is in equilibrium will fall”. [NB: in your answer, make sure to mention if, in the diagram depicting the equilibrium in the money market, the change in c will lead to a shift in the money supply curve, in the money demand curve, or in both, and why].

SKETCH OF ANSWER

True. The fall in c will rise the money multiplier and hence, for any given monetary base H , money supply M . In fact, if c goes down deposits will rise and, for any given reserve/deposits ratio θ , bank loans will go up, causing a further increase in deposits and therefore in money supply. In the diagram depicting the equilibrium in the money market, the money supply curve will shift to the right, crossing the money demand curve for a lower equilibrium interest rate.

Question 3 (7.5 points)

In the closed economy with constant prices Macro, the consumption function is $C = c_0 + c_1(Y - \bar{T}) + a\overline{WFH}$, where WFH is the individuals' financial and housing wealth, assumed to be exogenous, and the positive parameter a is smaller than the marginal propensity to consume, $0 < a < c_1$. Macro is described by an otherwise standard IS-LM model, with $I = \bar{I} + d_1Y - d_2i$, $G = \bar{G}$, $T = \bar{T}$ and a central bank that chooses the interest rate. Starting from an initial equilibrium position, individuals' wealth and net taxes fall at the same time and by the same amount, so that $\Delta\overline{WFH} = \Delta\bar{T} < 0$. To prevent these changes from affecting equilibrium income, the central bank decides to resort to an open market operation. To keep income constant, which kind of open market operation should the central bank implement? Why? Finally, discuss how and why the composition of the aggregate demand for goods will change in the move from the initial equilibrium to the new equilibrium that Macro will reach following the changes in $\overline{WFH}$ and in $\bar{T}$ and the monetary policy intervention you have suggested. Explain.

SKETCH OF ANSWER

If $\Delta\overline{WFH} = \Delta\bar{T} < 0$, autonomous spending rises by $(a - c_1)\Delta\bar{T}$ (a positive quantity, since $a < c_1$ and $\Delta\bar{T} < 0$), the IS curve shifts to the right, Y tends to go up and, to keep it unchanged, the central bank must shift the LM curve upwards. To do this through an open market operation, it must sell bonds, thus lowering their price and raising their yield (i).

In the new equilibrium, G is of course unchanged, while investment is lower (Y is unchanged, and i higher). It follows that consumption will have gone up by the same amount by which investment has fallen. In fact, since the supply of goods is kept at the initial level by the central bank, the demand for goods cannot change, either. This means that, in equilibrium, the effect on C of the decrease in net taxes, that tends to raise consumption, must prevail on that of lower wealth, a change that tends to lower it.

Question 4 (7.5 points)

Beta is an economy with flexible prices, described by an IS-LM-PC model with $\pi^e = \bar{\pi}$ and initially in a medium-run equilibrium position. Firms produce goods using a technology summarized by the production function $Y = AN$, where A is labour productivity and N is employment. Suppose now that, due to a technological innovation, A rises ($\Delta A > 0$). At the same time, due to a permanent increase in financial markets participants' degree of risk aversion, Beta also experiences a permanent rise in the risk premium, x . Having discussed which of the curves represented in the IS-LM-PC diagram will be affected by these two changes, and how, explain how the natural real rate of interest, r_n , will change in the new medium-run equilibrium that will now prevail in Beta, providing the economic intuition behind the change in r_n that you think will have to take place [NB: to provide this intuition, make explicit reference to the way in which the two changes under consideration will affect the two curves in the "supply of goods-demand for goods" diagram used to determine the equilibrium of the economy in the medium run and the associated level of the natural real interest rate r_n].

SKETCH OF ANSWER

In the WS-PS model, an increase in labour productivity shifts the PS curve up, thus leading to a decrease in the natural rate of unemployment. The ensuing increase in the natural level of production implies that, in the IS-LM-PC diagram, the PC curve shifts to the right, crossing the x-axis for the new, higher level of Y_n . By making the borrowing rate $r + x$ higher for any given r , the increase in x will instead shift the IS curve to the left. The natural real interest rate – the value of r for which the demand for goods is equal to Y_n – is now lower. In fact, and for given r_n , in the medium-run the two shocks under consideration lead to an excess supply of goods (for any given real interest rate, the medium-run supply of goods has gone up, the demand down). It follows that the natural interest rate will have to fall by the amount needed to eliminate this excess supply and make sure a new medium-run equilibrium will be reached.